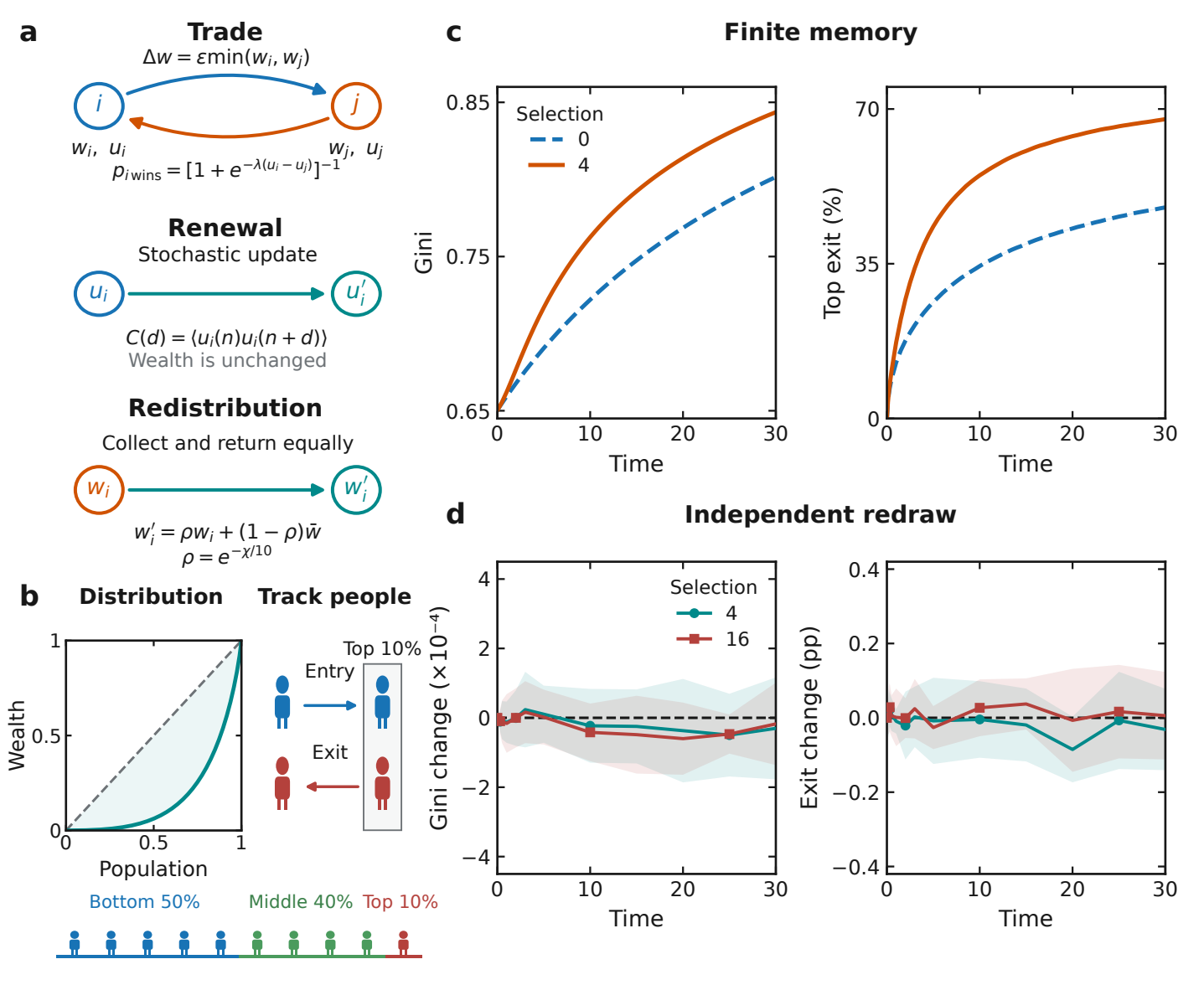
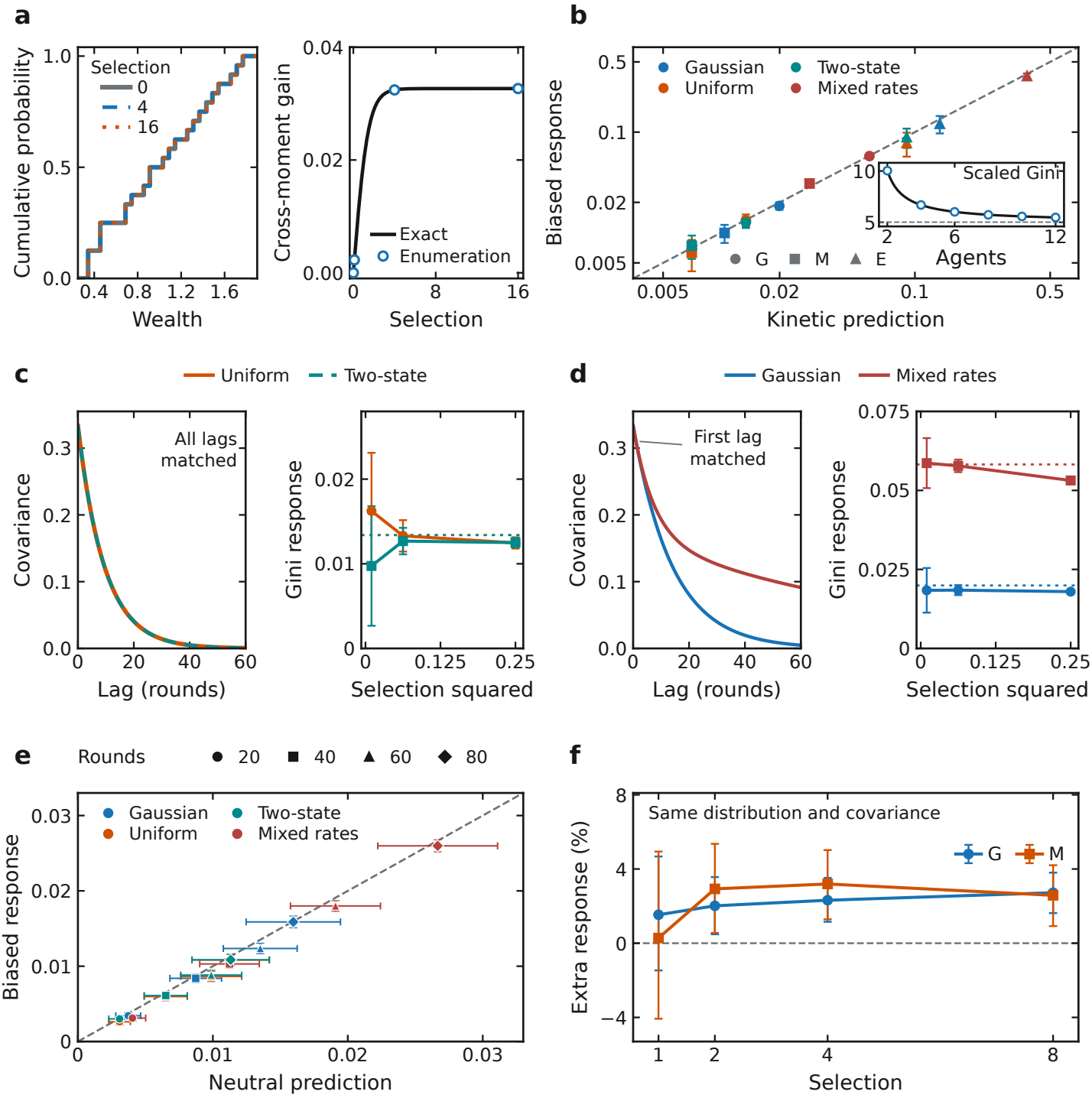
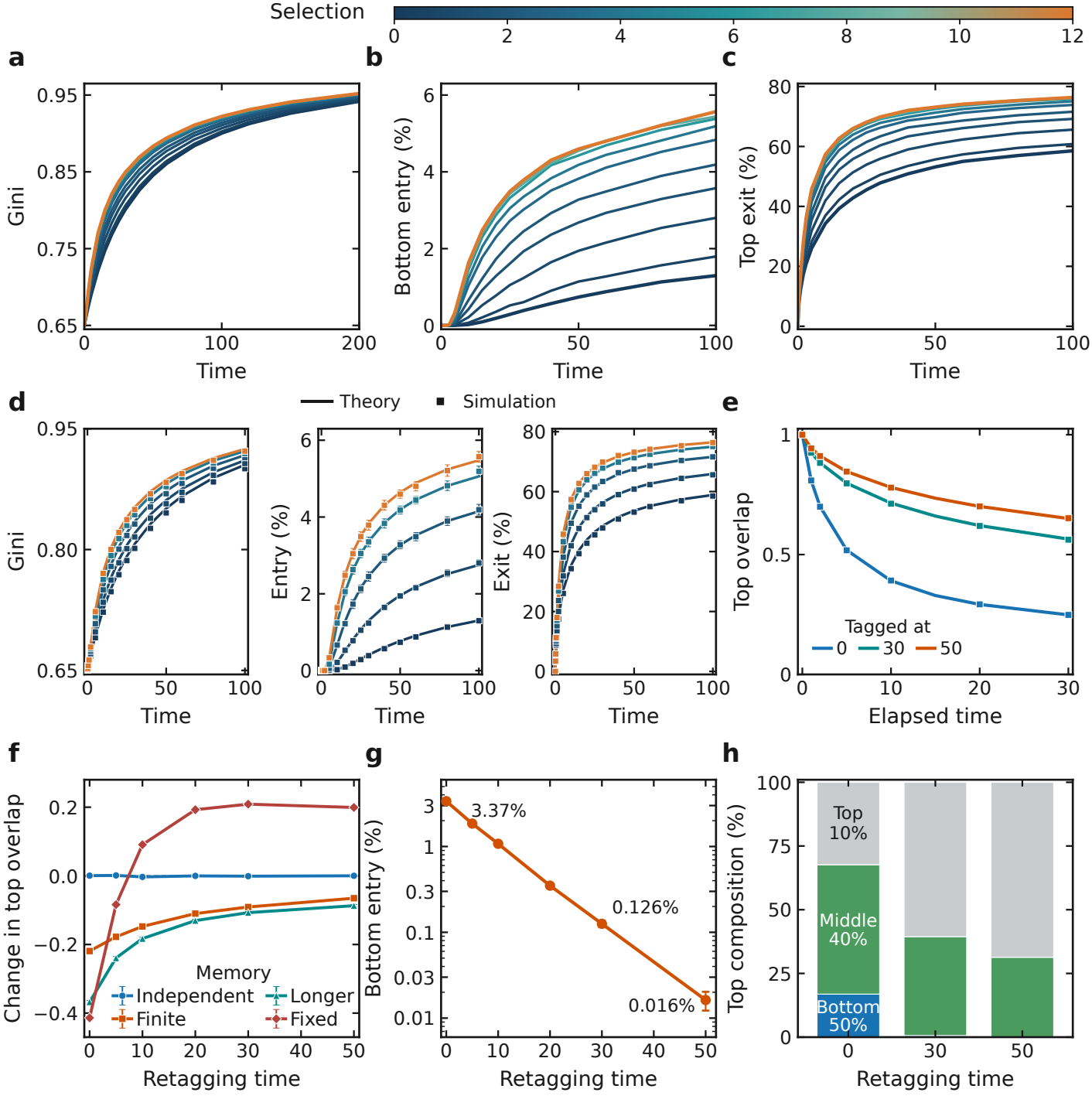
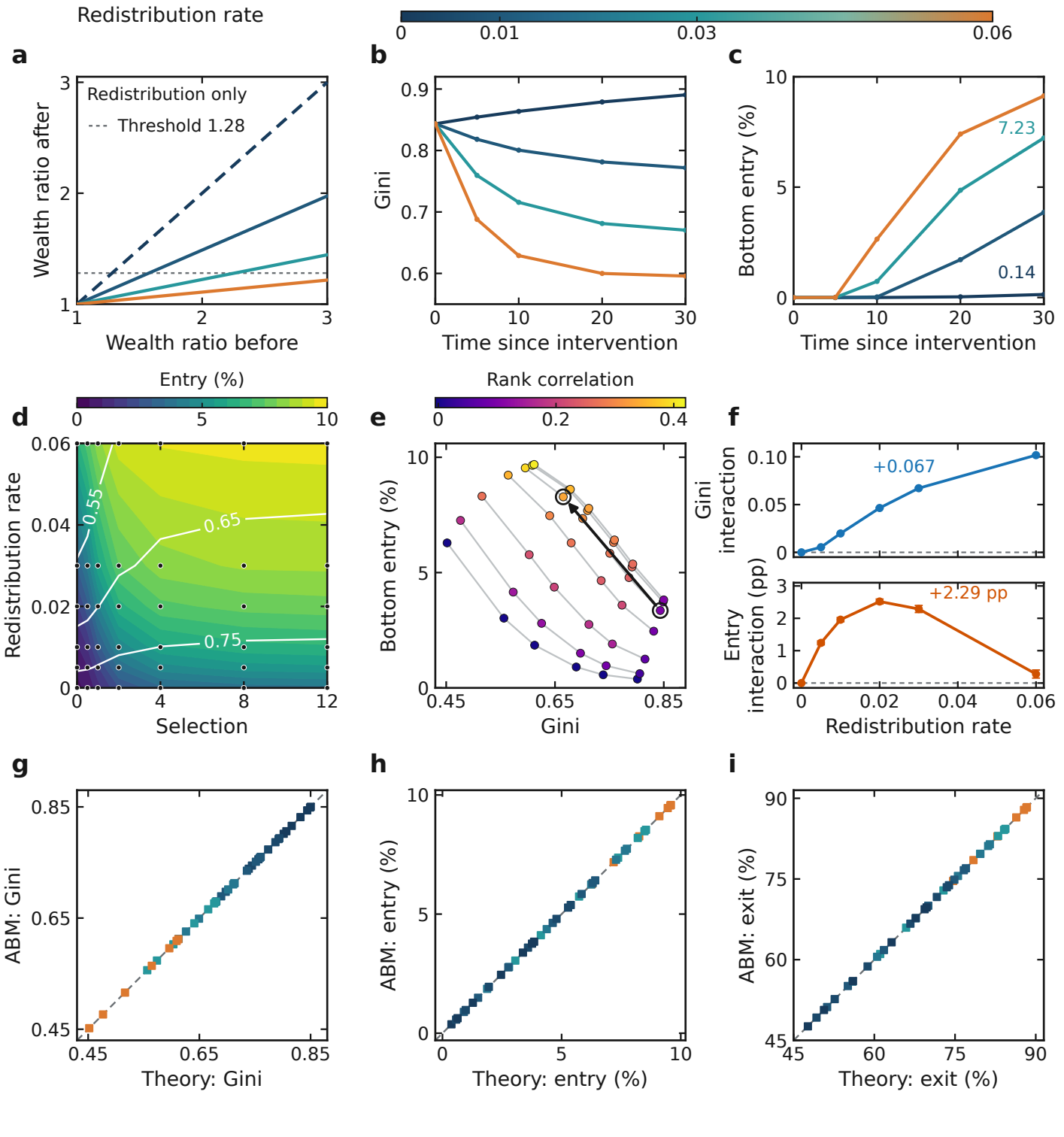

a
Trade
$\Delta w = \varepsilon \min(w_i, w_j)$
i
j
$w_i,\ u_i$
$w_j,\ u_j$
$p_{i\,\text{wins}} = [1 + e^{-\lambda(u_i - u_j)}]^{-1}$
Renewal
Stochastic update
u_i
u_i'
$C(d) = \langle u_i(n) u_i(n+d) \rangle$
Wealth is unchanged
Redistribution
Collect and return equally
w_i
w_i'
$w_i' = \rho w_i + (1-\rho)\bar{w}$
$\rho = e^{-\chi/10}$
b
Distribution
Track people
Wealth
Population
Top 10%
Entry
Exit
Bottom 50%
Middle 40%
Top 10%
c
Finite memory
Selection
0
4
Gini
Time
Top exit (%)
Time
d
Independent redraw
Selection
4
16
Gini change ($\times 10^{-4}$)
Time
Exit change (pp)
Time


a
Cumulative probability
Selection
0
4
16
Wealth
Cross-moment gain
Exact
Enumeration
Selection
b
Gaussian
Two-state
Uniform
Mixed rates
Biased response
Scaled Gini
Agents
G
M
E
Kinetic prediction
c
Uniform
Two-state
All lags matched
Covariance
Lag (rounds)
Gini response
Selection squared
d
Gaussian
Mixed rates
First lag matched
Covariance
Lag (rounds)
Gini response
Selection squared
e
Rounds
20
40
60
80
Gaussian
Two-state
Uniform
Mixed rates
Biased response
Neutral prediction
f
Same distribution and covariance
G
M
Extra response (%)
Selection


Selection
0
2
4
6
8
10
12
a
Gini
0.95
0.85
0.75
0.65
0
100
200
Time
b
Bottom entry (%)
6
4
2
0
0
50
100
Time
c
Top exit (%)
80
60
40
20
0
0
50
100
Time
d
Theory
Simulation
Gini
0.95
0.80
0.65
0
50
100
Time
Entry (%)
6
4
2
0
0
50
100
Time
Exit (%)
80
60
40
20
0
0
50
100
Time
e
Top overlap
1
0.5
0
Tagged at
0
30
50
0
10
20
30
Elapsed time
f
Change in top overlap
0.2
0.0
−0.2
−0.4
Memory
Independent
Longer
Finite
Fixed
0
10
20
30
40
50
Retagging time
g
Bottom entry (%)
3
1
0.3
0.1
0.03
0.01
3.37%
0.126%
0.016%
0
10
20
30
40
50
Retagging time
h
Top composition (%)
100
75
50
25
0
Top
10%
Middle
40%
Bottom
50%
0
30
50
Retagging time


Redistribution rate
0
0.01
0.03
0.06
a
Redistribution only
Threshold 1.28
Wealth ratio after
Wealth ratio before
b
Gini
Time since intervention
c
Bottom entry (%)
7.23
0.14
Time since intervention
Entry (%)
d
Redistribution rate
0.55
0.65
0.75
Selection
Rank correlation
e
Bottom entry (%)
Gini
f
Gini interaction
+0.067
Entry interaction (pp)
+2.29 pp
Redistribution rate
g
ABM: Gini
Theory: Gini
h
ABM: entry (%)
Theory: entry (%)
i
ABM: exit (%)
Theory: exit (%)